

MANIKANDA RAO A S

Senior Edge AI Engineer | Computer Vision • Intelligent Transport Systems • Smart City

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PROFESSIONAL SUMMARY

Edge AI and embedded systems engineer with 5+ years of hands-on experience building real-time computer vision and traffic analytics systems on NVIDIA Jetson platforms. Trains custom AI models on project-specific datasets and takes them through the full pipeline — data collection and annotation, training, TensorRT optimization, edge deployment, and backend delivery. Delivers Intelligent Transport Systems (ITS) and Smart City deployments across highways, toll plazas, and city traffic policing, with proven live results including 98% vehicle-classification accuracy on a system approved by the Government of Gujarat.

CORE COMPETENCIES

Domain: Intelligent Transport Systems (ITS), Smart City, ATCC, ANPR, MLFF Tolling, FASTag/RFID, Video Incident Detection

Edge AI Platforms: NVIDIA Jetson Orin AGX, Orin NX, Orin Nano, Xavier NX, Nano Series, Raspberry Pi

AI / Deep Learning: YOLOv8, YOLO11, YOLO26, NVIDIA DeepStream SDK, TensorRT, PyTorch, TensorFlow, Object Detection, Classification, ANPR

AI Model Development: Custom AI Model Training, Dataset Collection & Annotation, Data Augmentation, Transfer Learning, Fine-Tuning, Hyperparameter Tuning, AI Model Accuracy Validation (mAP, Precision/Recall), AI Model Optimization & Deployment, INT8 / FP16 / FP32 Quantization, ONNX & TensorRT Export

Programming: Python, C, Shell Scripting

Video & Computer Vision: GStreamer, FFmpeg, OpenCV, IP Cameras, RTSP, Multi-Stream Video Analytics

Backend & APIs: Python, FastAPI, REST APIs, Jinja2 Templating, NGINX

Databases: PostgreSQL

DevOps & Deployment: Docker, Jenkins, Git, CI/CD, Automated Service Startup

Cloud: AWS (EC2, S3, IAM, CloudWatch, Kinesis Streams), Hostinger VPS

OS & Networking: Linux/Ubuntu (x86 & ARM), Windows, TCP/UDP, FTP, HTTPS, Wired/Wireless Networking

Web & Remote Access: VNC Server, SSH & Tunnelling, DNS, SSL

PROFESSIONAL EXPERIENCE

COS ARTIFICIAL INTELLIGENCE PVT LTD

Madurai, Tamil Nadu | May 2021 – Present

Senior Edge AI Engineer

2024 – Present

- Lead delivery of real-time AI and computer vision systems for Intelligent Transport Systems and Smart City projects on NVIDIA Jetson Orin AGX, Orin NX, and Xavier NX platforms.
- Own end-to-end custom AI model development on YOLOv8, YOLO11, and YOLO26 — dataset collection, annotation, training, and accuracy validation — then deploy to edge devices for real-time inference.
- Convert trained AI models to TensorRT engines using INT8, FP16, and FP32 precision, tuning quantization and throughput to balance accuracy and latency.
- Lead production rollout, automated service startup, and cloud administration using Docker, Jenkins, Git, CI/CD, AWS (EC2, S3, IAM, CloudWatch), and Hostinger VPS.

Edge AI Engineer

2023 – 2024

- Built multi-stream video analytics pipelines using NVIDIA DeepStream SDK, GStreamer, OpenCV, and FFmpeg with IP camera and RTSP inputs.
- Trained and fine-tuned object detection and classification models in PyTorch for vehicle detection, counting, and vehicle-type classification on live traffic footage.
- Integrated AI models with backend services using Python, FastAPI, Jinja2, and PostgreSQL, delivering REST APIs and web dashboards for real-time traffic data.

Embedded Systems Engineer

May 2021 – 2023

- Set up and maintained NVIDIA Jetson and Raspberry Pi edge hardware, covering JetPack flashing, Linux configuration, and camera commissioning.
- Developed embedded software in Python, C, and shell scripting, including automated service startup for unattended field units.
- Integrated IP cameras, RFID readers, and field networking over TCP/UDP, FTP, and HTTPS, and resolved Jetson, Linux, and video-streaming issues in production.

OSIZ TECHNOLOGIES

Madurai, Tamil Nadu | 2018 – 2019

MEAN Stack Developer

- Developed web applications on the MEAN stack (MongoDB, Express.js, Angular, Node.js) for client projects.
- Built REST APIs, database schemas, and responsive front-end interfaces, working with design and QA teams through the full development cycle.

KEY PROJECTS**MLFF — Multi-Lane Free Flow Tolling**

Aniyari Toll Plaza, Gujarat

Client: Ahmedabad Toll Private Limited (AMTPL) — a Sekura Roadways Pvt Ltd company

Edge AI tolling platform running real-time vehicle analytics across four synchronized cameras, with on-site AI model fine-tuning for live traffic conditions. Integrated an RFID reader to read FASTag tag IDs and user vehicle codes in real time and matched them against ANPR plate reads, reaching **96% matching accuracy**. Owned Edge AI software development, deployment, camera and video pipeline integration, AI inference, troubleshooting, and production optimization.

ATCC — Vehicle Detection & Classification

Rajkot-Vadinar Highway, Gujarat

Client: Rajkot Vadinar Tollway Limited (RVTL) — a Sekura Roadways Pvt Ltd company

Highway ITS system for real-time vehicle detection, counting, and multi-class classification, achieving **98% classification accuracy** and approved by the Government of Gujarat. Running continuously in production for **3+ years** to date. Responsible for custom AI model training and TensorRT optimization, edge deployment, camera and video pipeline integration, accuracy validation, and ongoing production support.

VIDES — Video Incident Detection and Enforcement System (R&D)

Salem Highway, Tamil Nadu

Client: In-house research and development — COS Artificial Intelligence Pvt Ltd

Research and development of a real-time video incident detection and automated enforcement system on NVIDIA Jetson, covering accidents, violations, congestion, and vehicle-type analysis. Built DeepStream-based multi-camera analytics with integrated reporting, and validated detection performance on live highway footage.

ATCC — Automatic Traffic Counting and Classification (Fixed & Movable Kits)

Gujarat & Tamil Nadu

Client: L&T IDPL – Malvan Toll Plaza, Gujarat | Greater Chennai Traffic Police, Tamil Nadu

Vision-based traffic analytics delivered in two form factors. Fixed kits at National Highway toll locations analyze traffic flow, count vehicles, classify vehicle types, and publish real-time traffic data. The portable movable kit supports Smart City traffic policing with vehicle movement tracking and classification, combining edge computing, real-time data generation, embedded software, and a built-in web server for data access and analysis.

EDUCATION**M.E., Embedded Systems Technologies — Solamalai College of Engineering, Anna University**

2019 – 2021

First Class. Coursework: embedded hardware and software design, IoT applications.

B.E., Electronics and Communication Engineering — KLN College of Engineering, Anna University

2015 – 2018

First Class (65%).

Diploma, Electronics and Communication Engineering — KLN Polytechnic College

2012 – 2015

Passed with Distinction.

LANGUAGES & INTERESTS**Languages:** English — Professional proficiency | Tamil — Native proficiency**Interests:** Cricket; marathon running and competitive road races